Randy Tang

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SUMMARY

Software engineer with 4+ years of experience designing and optimizing high-throughput distributed systems and data pipelines processing tens of millions of daily transactions. Strong systems-level thinking (Cornell BS in CS and Mechanical Engineering), deep Python/Postgres proficiency, and hands-on experience building robust monitoring, alerting, and parallelized data frameworks. U.S. citizen willing to relocate to Littleton, CO.

TECHNICAL SKILLS

Languages: Python, SQL, Java (fundamentals/coursework) | **Data/Systems:** Postgres, SQLite, Pandas, AMPS message queues, Docker, high-throughput pipelines, concurrency, telemetry, microservices
Web/Frameworks: Python web frameworks (Flask), Grafana, REST APIs | **Practices:** Distributed systems architecture, performance optimization, CI/CD, Git, code review, documentation

EXPERIENCE

Software Engineer — Bank of America
Aug 2022 – Present

Cirrus OTC Trade Reporting | Python, Pandas, Quartz, AMPS

- Designed, built, and optimized **high-throughput, production-critical distributed pipelines** reporting all OTC trades, processing **tens of millions of transactions daily**.
- Built an extensible **reference-data framework of microservices** connected via AMPS message queues querying multiple external sources, handling retries, load balancing, and failure mitigation.
- Implemented a **T+1 Unique Trade Identifier service** validating trade-ID uniqueness across counterparties and issuing corrections, processing millions of transactions per day.
- Developed a parallelized **high-volume backreporting service** to automate previously manual workflows, delivering substantial performance and throughput improvements.
- Designed and implemented a data transformer (internal model to outgoing XML) and a DTCC response parser to update transaction statuses dynamically.
- Partnered with upstream database/systems teams and downstream stakeholders to analyze 50–100GB datasets using Python/Pandas, optimizing system latency and data flows.

Data Analyst (Contract) — Microsoft
Mar 2022 – Aug 2022

Viva Topics | Python, Pandas, Jupyter

- Collaborated with the data science team on Viva Topics; wrote Python scripts to process, clean, and annotate user data used to train ML models.

Engineering Intern — Northrop Grumman
Fall 2019, Summer 2020

NX, ANSYS

- Designed and released 50+ aerospace engineering drawings including flight and test components; performed structural analyses for large assemblies and detailed fastener stacks.

PROJECTS

- **Market Data Dashboard** (Grafana, Postgres, Docker, Twilio) — Ingests OI, funding, and other market data from Binance and Hyperliquid APIs; designs schema and sets up dashboards plus automated SMS alerts on signal triggers.
- **Polymarket Scanner** (Flask, Discord API, SQLite) — Scans Polymarket for large trades and unusual activity; implements a lightweight backend database and pushes real-time alerts to Discord.
- **Reading Recommendations App** (Flask, Gemini API, SQLite) — Implemented an LLM-powered daily recommender over a Project Gutenberg corpus.

EDUCATION

Cornell University, College of Engineering — Ithaca, NY
Sep 2017 – Dec 2021

B.S. Computer Science & B.S. Mechanical Engineering | GPA 3.7

- **CUAIR (Cornell Unmanned Aircraft)** — Developed systems for an autonomous plane including obstacle avoidance, object classification, and airdrop; top-performing team at the AUVSI SUAS competition.